



Lightweight Machine Learning for Edge-Based Machinery Fault Diagnosis

Manh-Cuong Nguyen¹, Quoc-Chien Truong¹, Phuc-Tan Le^{1,2}, Ngoc-Lam Ngo¹,
Phong-Dien Nguyen¹, and Trong-Du Nguyen¹(✉)

¹ School of Mechanical Engineering, Hanoi University of Science and Technology,
Hanoi, Vietnam

`du.nguyentrong@hust.edu.vn`

² Department of Mechatronics and Automotive Engineering, Hanoi University of
Business and Technology, Hanoi, Vietnam

Abstract. Prognostics and Health Management (PHM) are crucial to ensuring the reliability and safety of production systems and machines. However, the widespread implementation of these technologies in practical systems remains challenging, especially in cost-sensitive environments. This study proposes a lightweight fault-diagnosis framework that focuses on frequency-domain feature extraction to minimise the memory footprint and optimise execution speed. Leveraging these features, optimised classical machine learning models are implemented independently of heavyweight libraries, facilitating seamless integration onto hardware-limited hardware. Experimental results show that the Random Forest model, utilising a minimal number of decision trees, can achieve acceptable classification performance and effectively avoid overfitting. Besides that, Softmax Regression achieves comparable accuracy while converging better. This research underscores that the combination of essential signal processing and lightweight machine learning provides an efficient and reliable solution for edge-based machinery fault diagnosis.

Keywords: Edge-Computing · Machinery fault diagnosis ·
Lightweight machine learning · Vibration signal analysis

1 Introduction

Reliability and safety are crucial for mechanical systems in today's industries. In the past, experts relied on expert-based systems, model-based methods, and classical signal processing. However, these conventional methods face several challenges in modern settings. Expert-based systems depend on subjective judgment and do not scale well [1, 2]. Model-based methods struggle to adapt because building accurate mathematical models is complex [3–5]. Classical frequency analysis [6] is easily affected by noise and changes in load, and often requires careful parameter/threshold selection [7–9]. Machine learning (ML) now offers

a new way to automatically classify faults using past data, without needing detailed physical models [10]. Still, there is a research gap because most AI systems [11] rely on cloud-based setups. Sending high-frequency sensor data to the cloud causes delays, uses more bandwidth [12]. These issues make cloud-based systems a poor fit for real-time, safety-critical industrial uses. To address the need for decentralised solutions, this article presents a lightweight fault-diagnosis framework designed for affordable edge devices. It uses advanced signal processing and improves classic ML algorithms, such as Random Forests, to balance accuracy and efficiency. This approach allows reliable, real-time fault diagnosis at the edge, without relying on cloud systems that may not always be stable. The remainder of this paper is structured as follows: Sect. 2 presents the frequency-domain signal processing method and the Random Forest model architecture. Next, Sect. 3 discusses the experimental setup, including the datasets description, Random Forest HyperParameters, and the impact of model optimization computational efficiency. Finally, Sect. 4 concludes the paper, summarizing the main contributions and outlining future research directions.

2 Methodology

2.1 Proposed Framework

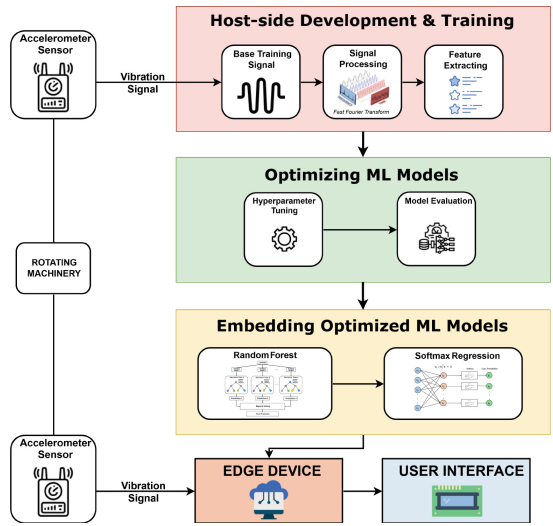


Fig. 1. Workflow of the Host-side model optimizing and training

The overall workflow of the fault diagnosis framework is illustrated in Fig. 1. Vibration signals acquired from an accelerometer mounted on the rotating machine are used as the primary input, since vibration responses are highly

sensitive to mechanical faults and variations in operating conditions. In host-side development, vibration signals are converted to the frequency domain to highlight key fault-related patterns. To keep computation and memory use low, only the most important frequency-domain features are used, and advanced time-frequency or adaptive signal-processing methods are left out. This set of features gives enough information to classify faults and works well for embedded systems. Using these features, two lightweight machine learning models, Random Forest and Softmax Regression, are trained and fine-tuned. Random Forest handles complex, nonlinear links between features and faults and is robust, while Softmax Regression is a fast, stable linear classifier. Model settings are adjusted to find a good balance between accuracy and efficiency. Once optimised, the feature extraction and trained models create a compact diagnostic system that fits well in edge devices with limited resources. This setup enables quick, on-device fault detection without relying on cloud computing.

2.2 Frequency-Domain Analysis

The theoretical and mathematical foundations in this section are adapted from Nguyen (2023) [6], which provides a systematic treatment of vibration measurement and analysis. The Fourier transform, derived from Fourier series theory, is central to vibration analysis, expressing any periodic function $x(t)$ with period T as a sum of harmonic components.

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j2\pi f_k t} \quad (1)$$

where j denotes the imaginary unit and X_k are the complex coefficients.

For digital signal processing, a signal $x(n)$ with N sampled points is analyzed using the Discrete Fourier Transform (DFT), defined as:

$$X_k = \frac{1}{N} \sum_{n=1}^N x(n) e^{-j \frac{2\pi(k-1)(n-1)}{N}} \quad (2)$$

where $k = 1, 2, \dots, N$. The amplitude-frequency spectrum $A(f_k)$ is subsequently calculated from the complex-valued DFT results:

$$A(f_k) = 2|X_k| \quad (3)$$

In cases where the number of sampled points N is a power of 2 ($N = 2^m$), the Discrete Fourier Transform can be computed very efficiently using the Fast Fourier Transform (FFT) algorithm. Ten FFT-based features are extracted from each vibration segment: mean frequency, RMS frequency, variance, skewness, kurtosis, central frequency, standard deviation, spectral centroid, spectral spread, and spectral entropy. These features jointly capture the location, concentration, shape, and complexity of spectral energy, allowing the model to distinguish fault-related peaks from healthy conditions with low computational and memory cost, which is well suited for edge implementation.

2.3 Data Processing Protocol and Dataset Splitting

The raw vibration signals are first segmented into non-overlapping windows of 8192 samples, corresponding to $\frac{8192}{25600} \approx 0.32$ s per segment at the sampling rate of 25.6 kHz. The window length is selected as a power of two to fully exploit the computational efficiency of the FFT algorithm, while remaining short enough to limit memory usage on edge hardware. For single-precision representation, each window requires approximately 8192×4 bytes per channel (about 32 kB for two channels), which fits comfortably within the RAM constraints of typical low-cost microcontrollers. Non-overlapping windows are adopted to reduce computational load and simplify buffer management for real-time edge deployment. While overlapping can improve sensitivity to short transients, the vibration signals in this study are collected under steady-speed conditions, where fault signatures are predominantly periodic. Therefore, non-overlapping segmentation remains effective without significantly affecting diagnostic performance. All available segments are then randomly shuffled and partitioned into three disjoint subsets following a 70/15/15 ratio for training, validation, and testing, respectively, yielding a total of 26,112,000 labeled samples across all machine conditions and load levels. Random splitting is applied at the segment level so that each subset contains a representative mixture of normal, unbalance, misalignment, inner race fault, and outer race fault data under different torques. The training set is used to fit the model parameters, the validation set guides hyperparameter tuning and early stopping, and the held-out test set provides an unbiased estimate of the final diagnostic performance. This protocol ensures that the reported results reflect the generalization capability of the proposed lightweight framework, rather than overfitting to a specific subset of the vibration recordings.

2.4 Random Forest Model Architecture

Random Forest (RF)[13] is a robust ensemble learning algorithm that constructs a multitude of independent decision trees to solve classification and regression problems. In rotating machinery fault diagnosis, RF is selected for its ability to handle complex non-linear relationships and its stability against noisy data. The fundamental principle of RF relies on two key techniques: bootstrap sampling (bagging) and random feature selection. Each decision tree in the forest is trained on a bootstrap sample X_b drawn randomly with replacement from N original training instances (Fig. 2):

$$X_b = \{x_{i1}, x_{i2}, \dots, x_{iN}\}, i_k = \{1, \dots, N\} \quad (4)$$

During tree construction, at every internal node, the dataset S is divided into two disjoint subsets S_L and S_R based on a selected feature f and a corresponding threshold t :

$$S_L = \{x_i, y_i | x_{i,f} \leq t\}, S_R = \{x_i, y_i | x_{i,f} > t\} \quad (5)$$

To evaluate the quality of a candidate split, the Gini impurity measure is employed:

$$Gini(S) = 1 - \sum p_c^2 \quad (6)$$

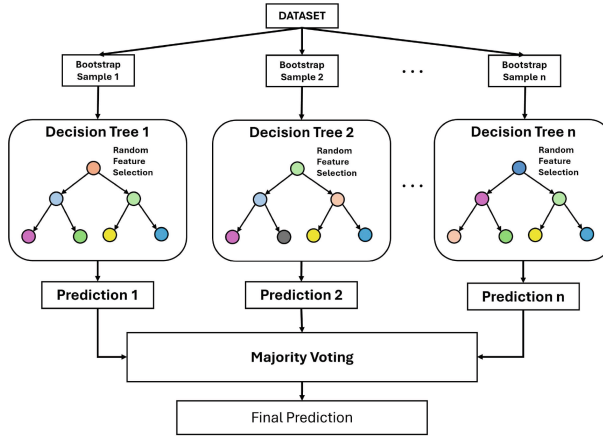


Fig. 2. Random forest algorithm

where p_c is the proportion of class c in subset S . The optimal split is chosen by minimizing the weighted Gini index:

$$Gini_{split} = \frac{|S_L|}{|S|} Gini(S_L) + \frac{|S_R|}{|S|} Gini(S_R) \quad (7)$$

Let $T_1, T_2, \dots, T_M$ denote the set of decision trees in the forest. For a given input sample x , each tree produces a class prediction:

$$\hat{y}_i = T_i(x) \quad (8)$$

The final classification result is obtained by aggregating the predictions from M individual trees through a majority voting mechanism:

$$y = \operatorname{argmax} \sum_{i=1}^M I(y_i = c) \quad (9)$$

3 Experiment Results

3.1 Datasets Description

Dataset: *Vibration, Acoustic, Temperature, and Motor Current Dataset of Rotating Machine Under Varying Load Conditions for Fault Diagnosis* [14].

The Vibration, Acoustic, Temperature, and Motor Current Dataset was collected from a rotating-machinery testbed under various fault conditions and load levels. The dataset covers normal operation, bearing inner-, outer-race faults, shaft misalignment, and rotor unbalance at loads of 0 Nm, 2 Nm, and 4 Nm. The testbed includes a three-phase, four-pole induction motor (3 HP, SIEMENS), a gearbox, shafts, two bearing housings (A and B), rotors, and a hysteresis brake.

The system runs at 3010 rpm to avoid power-frequency interference. Several sensors are used, such as accelerometers, microphones, thermocouples, current sensors, and an RPM sensor. Four accelerometers are placed on bearing housings A and B in the x- and y-directions, following ISO 10816-1. While the dataset includes data from several sensors, this study uses only the vibration signals from Accelerometer A, mounted on Bearing Housing A, in the x and y directions. This spot is especially sensitive to bearing faults and mechanical transmission issues. The vibration dataset covers five machine conditions (see Table 1): normal operation, rotor unbalance, shaft misalignment, bearing inner race fault (IRF), and bearing outer race fault (ORF). The normal condition has 120 s of data, and each fault condition has 60 s. All data was recorded at a steady speed of 3010 rpm. Vibration signals were sampled at 25.6 kHz to capture fault-related frequencies and harmonics.

Table 1. Experimental Dataset Specifications[14]

Data type	Fault types	Length (second)	Rotating speed (RPM)	Sampling rate (kHz)
Vibration	Normal	120	3010	25.6
	Unbalance	60	3010	25.6
	Misalignment	60	3010	25.6
	Inner Race Fault	60	3010	25.6
	Outer Race Fault	60	3010	25.6

3.2 Random Forest HyperParameters

Random Forest HyperParameters were optimised to achieve high diagnostic accuracy while maintaining low computational complexity, suitable for embedded implementation. Key hyperparameters, including the number of trees (Fig. 3), maximum tree depth (Fig. 4), and maximum number of leaf nodes (Fig. 5), were systematically tuned based on validation accuracy. As indi-

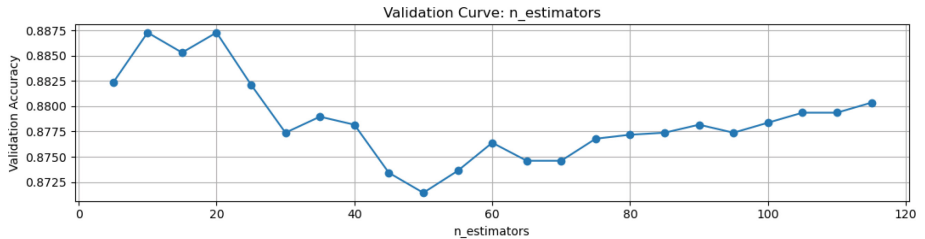


Fig. 3. Validation curve showing the effect of the number of trees on validation accuracy.

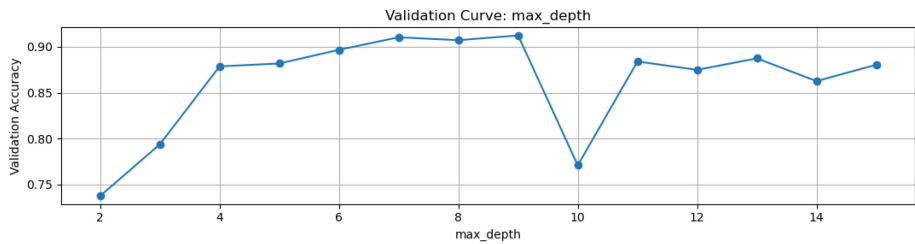


Fig. 4. Validation curve showing the effect of the maximum tree depth on validation accuracy.

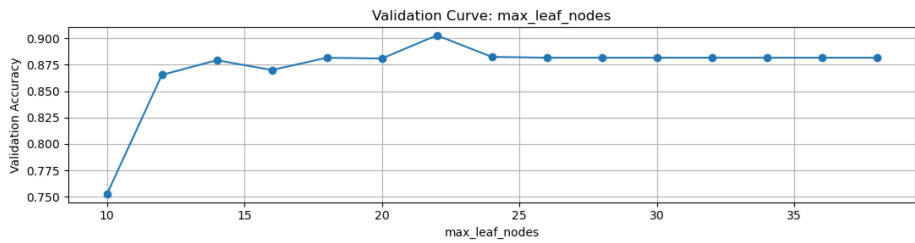


Fig. 5. Validation curve showing the effect of the maximum number of leaf nodes on validation accuracy.

cated by the results, increasing the number of trees provides diminishing validation accuracy gains, leading to the selection of a small ensemble size to reduce computational cost. An intermediate tree depth was chosen to balance model complexity and generalization, resulting in hyperparameters suitable for reliable fault classification and efficient real time inference on embedded systems (Table 2).

Table 2. Training Hyperparameters of Random Forest

Hyperparameter	Search Range	Selected
n_estimators	(10, 120)	10
max_depth	(2, 16)	7
max_leaf_nodes	(10, 40)	22

3.3 Impact of Model Optimization Computational Efficiency

The confusion matrix in Fig. 6 shows that the Random Forest model achieves high accuracy (99.52%), correctly classifying most fault types, with only minor

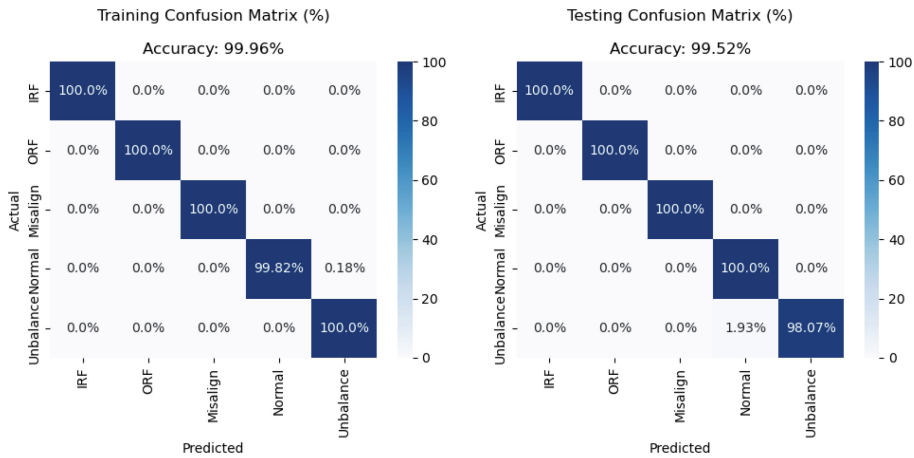


Fig. 6. Random Forest Confusion Matrix

confusion between Unbalanced and Normal (1.93%), likely due to similar spectral characteristics. The close agreement between training (99.96%) and testing accuracy indicates good generalisation.

As shown in Fig. 7, the Random Forest model outperforms Softmax Regression, particularly for non-linear decision boundaries. While both models perform well on distinct faults (e.g., ORF, Misalignment), Softmax Regression struggles to separate Normal and Unbalanced ($F1 \approx 89\text{--}91\%$), whereas Random Forest maintains higher consistency ($F1 \approx 98\text{--}99\%$). The optimisation process led to strong performance improvements, with higher test accuracy and much lower computational overhead. As shown in Table 3 and Fig. 8, accuracy increased while both RAM use and inference latency dropped by over 50%. Softmax Regression still uses about 6 KB less RAM and is 5 ms faster than Random Forest, but this difference is minor for modern microcontrollers. The proposed framework targets an ESP32-based edge device (Xtensa LX6, up to 240 MHz) which achieves an estimated inference latency of 33.6 ms per window (Random Forest) and 28.6 ms per window (Softmax Regression) under this hardware constraint, supporting real-time deployment.

Table 3. Effect of Optimization on Model Performance

Model	Test Accuracy (%) RAM Usage (bytes) Inference latency (ms/window)					
	Pre-Opt.	Post-Opt.	Pre-Opt.	Post-Opt.	Pre-Opt.	Post-Opt.
Random Forest	85.7	99.52	412,758	194,380	67.1	33.6
Softmax Regression	78.8	95.69	387,695	187,951	59.8	28.6

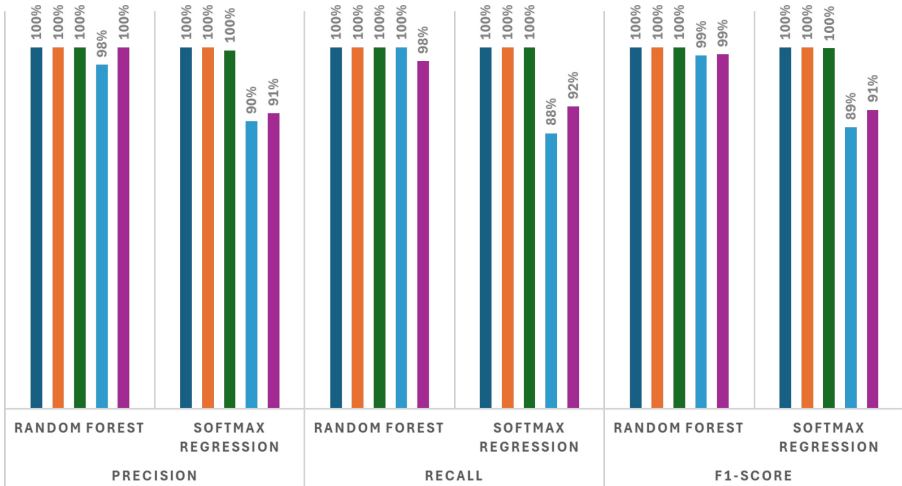


Fig. 7. Comparison of Classification Performance for Random Forest and Softmax Regression

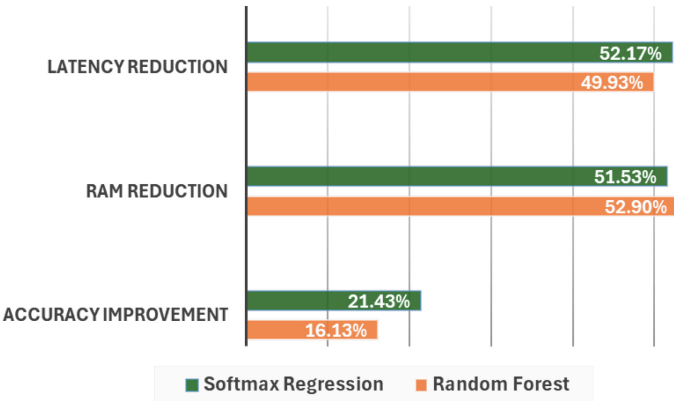


Fig. 8. Relative Performance Gains Achieved Through Model Optimization

4 Conclusions

This study introduces a fault-diagnosis deployment process for rotating machinery that works well on affordable edge devices. Instead of using complex signal processing or heavy models, it combines frequency-domain features with an optimised Random Forest classifier. This makes it possible to detect faults accurately while keeping computational and memory requirements low. Tests with a multi-fault vibration dataset show that the Random Forest model performs well, even when using a small number of trees. The optimisation improves classification accuracy and reduces memory use and response time. These findings show that the framework is well-suited for real-time embedded systems with limited hard-

ware. The results show that it is possible to diagnose machinery faults without using deep or complex models. Future research will focus on running the method directly on devices and testing it in a wider range of real-world conditions, such as changing speeds, varying loads, and real industrial noise.

References

1. White, M.F.: Expert systems for fault diagnosis of machinery. *Measurement* **9**(4), 163–171 (1991)
2. Liao, S.-H.: Expert system methodologies and applications—a decade review from 1995 to 2004. *Expert Syst. Appl.* **28**(1), 93–103 (2005)
3. Narasimhan, S., Vachhani, P., Rengaswamy, R.: New nonlinear residual feedback observer for fault diagnosis in nonlinear systems. *Automatica* **44**(9), 2222–2229 (2008)
4. Odendaal, H.M., Jones, T.: Actuator fault detection and isolation: An optimised parity space approach. *Control. Eng. Pract.* **26**, 222–232 (2014)
5. Mulumba, T., Afshari, A., Yan, K., Shen, W., Norford, L.K.: Robust model-based fault diagnosis for air handling units. *Energy Build.* **86**, 698–707 (2015)
6. Nguyen, P.D.: *Ky thuat do va phan tich dao dong co hoc*. Ha Noi: Nha xuất bản Giáo dục Việt Nam (2023)
7. Nguyen, T.-D., Dien, N.: Advanced signal decomposition methods for vibration diagnosis of rotating machines: a case study at variable speed, pp. 393–400 (2021)
8. Yan, R., Gao, R.X., Chen, X.: Wavelets for fault diagnosis of rotary machines: a review with applications. *Sig. Process.* **96**, 1–15 (2014). Time-frequency methods for condition based maintenance and modal analysis
9. Nguyen, T.-D., Dien, N., Nguyen, H.C.: Detection fault symptoms of rolling bearing based on enhancing collected transient vibration signals, pp. 373–384 (2022)
10. Lei, Y., Yang, B., Jiang, X., Jia, F., Li, N., Nandi, A.K.: Applications of machine learning to machine fault diagnosis: A review and roadmap. *Mech. Syst. Signal Process.* **138**, 106587 (2020)
11. Nguyen, T.-D., Nguyen, T.-H., Do, D.-T.-B., Pham, T.-H., Liang, J.-W., Nguyen, P.-D.: Efficient and explainable bearing condition monitoring with decision tree-based feature learning. *Machines* **13**(6), 467 (2025)
12. Dong, J., Li, Z., Zheng, Y., Luo, J., Zhang, M., Yang, X.: Real-time fault detection for iiot facilities using ga-att-lstm based on edge-cloud collaboration. *Front. Neurorobot.* **18** (2024)
13. Hastie, T., Tibshirani, R., Friedman, J.: *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. New York, NY, Springer, 2 ed., (2009)
14. Jung, W., Kim, S.-H., Yun, S.-H., Bae, J., Park, Y.-H.: Vibration, acoustic, temperature, and motor current dataset of rotating machine under varying operating conditions for fault diagnosis. *Data Brief* **48**, 109049 (2023)